Section Report 3 — Independent Amplicon Analysis

BIOL 4810

2026-04-16

Section Report 3 — Independent Amplicon Analysis

Goal: Use your **own sequencing dataset** to complete the full workflow in Galaxy: process amplicon reads with **DADA2**, then visualize your results in **phyloseq** using **alpha diversity**, **beta diversity**, and a **taxonomic bar plot**.
Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

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0) Your dataset

0A. Sample set / project name:

Bird Microbiome

0B. Sequencing type:

Paired-End Sequencing

0C. Metadata grouping variable you analyzed:

Population, Sex, & Flock

Workflow overview

For this section report, you will complete:

```
plotQualityProfile → filterAndTrim → learnErrors → dada → mergePairs →
makeSequenceTable → removeBimeraDenovo → assignTaxonomy → phyloseq alpha
diversity → phyloseq beta diversity (NMDS using Bray–Curtis) → phyloseq
taxonomic bar plot
```

Reminder: This section report asks you to apply the entire workflow to your own data and explain the biological meaning of each major output.

Part 1 — Quality profiles

1A) Forward read quality profile

What is the purpose of this step?

To make sure that our sample is clear of any contaminants. Offers a visual of the quality.

Upload screenshot: forward plotQualityProfile PDF

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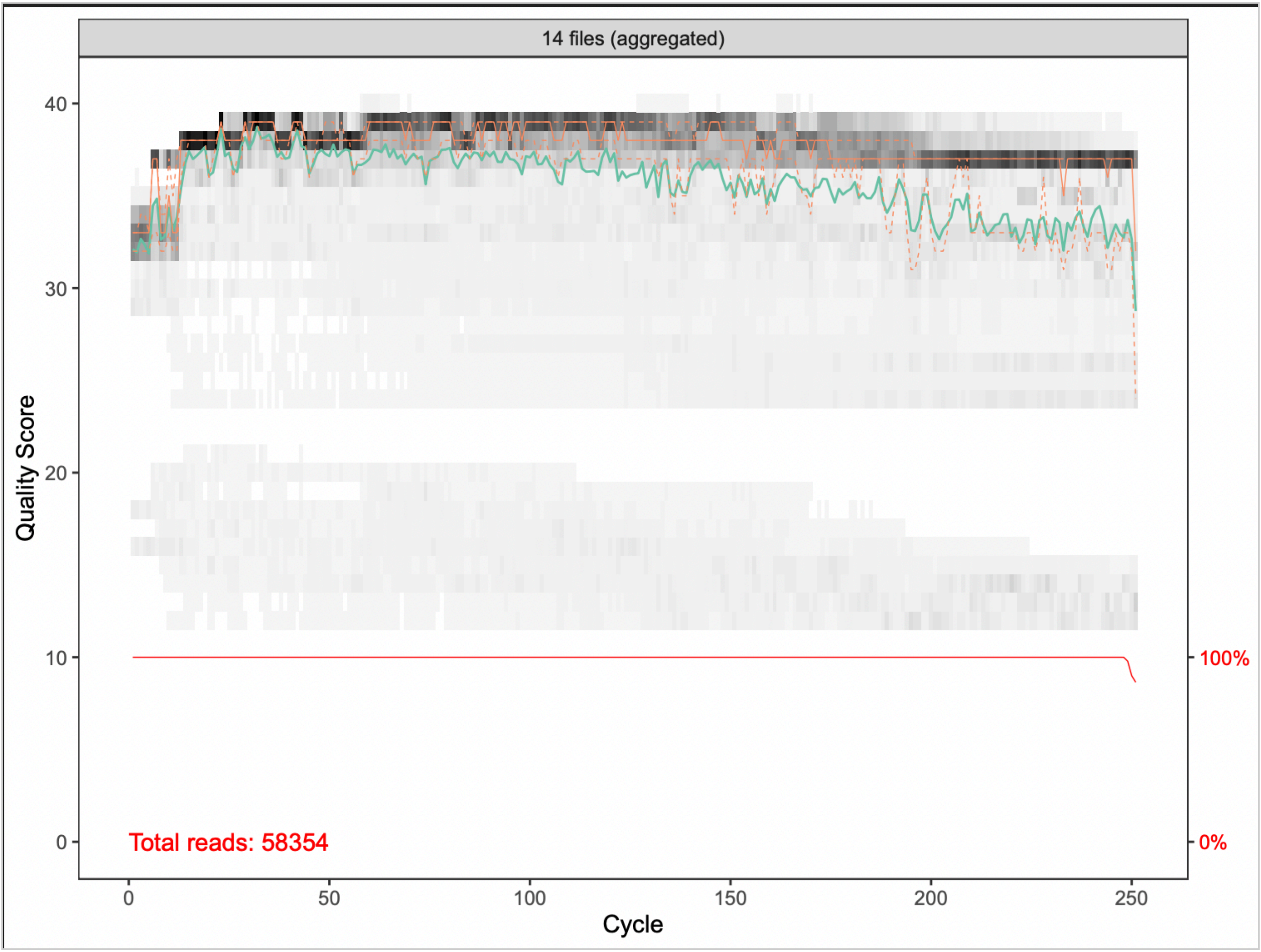

Where does quality begin to noticeably decline in the forward reads?

around 250

Chosen forward truncation length:

250

1B) Reverse read quality profile

What is usually different about reverse-read quality compared with forward-read quality?

The reverse read tends to degrade faster, meaning it drops below the 30 mark much sooner than the forward-read quality.

Upload screenshot: reverse plotQualityProfile PDF

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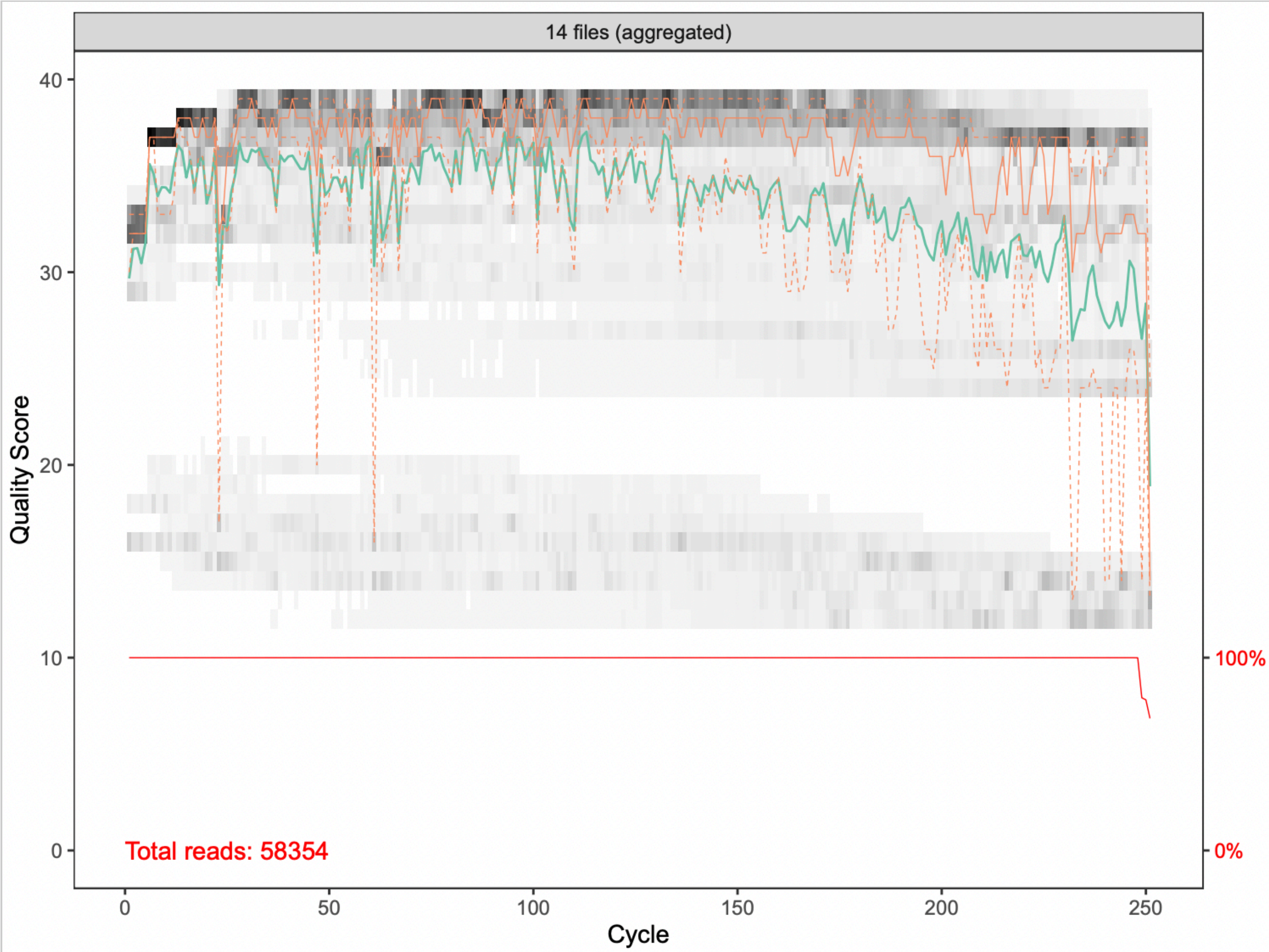

Where does quality begin to noticeably decline in the reverse reads?

Around 230

Chosen reverse truncation length:

230

1C) Truncation decision

Explain why you chose these truncation values.

Usually, most values over 30 are accurate. It's normally when values drop under 30 that the error rate begins to increase.

Rule of thumb: trim where quality begins to drop, but still keep enough overlap for forward and reverse reads to merge.

Part 2 — filterAndTrim

What filterAndTrim does:

This step removes low-quality reads and trims poor-quality bases from the ends of reads before denoising.

Why is this step important?

The Filter & Trim step cleans the data so that we're left with only high-quality pairs, allowing us to avoid errors.

Parameters used:

I only changed my truncate length, which was 250 forward and 230 reverse.

Upload screenshot: filterAndTrim output

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What could happen if you trim too aggressively?

We might not be able to make the sequences merge into a single sequence.

What could happen if you do not trim enough?

We could get incorrect results, leading to false assumptions.

IMPORTANT — Split Paired Reads (Unzip Collection)

After `filterAndTrim` , your output is a **paired collection.**
Before continuing, you must split this into **forward and reverse collections.**

What to do in Galaxy

1. In the Tools panel, search: **unzip collection**
2. Input: your `filterAndTrim` paired output
3. Run

What you will get

- One collection = **forward reads (R1)**
- One collection = **reverse reads (R2)**

Why this matters

`learnErrors` and `dada` must be run **separately** on forward and reverse reads.
If you skip this, `mergePairs` will fail.

Part 3 — learnErrors

What learnErrors does:
This step models the sequencing error rates in your dataset so DADA2 can distinguish real sequence variants from sequencing mistakes.

Why is this step important?

Learn Errors helps our program learn the "quirks" of our machine. It's important so that we can avoid the mistakes of misidentification of species.

Upload screenshot: learnErrors PDF

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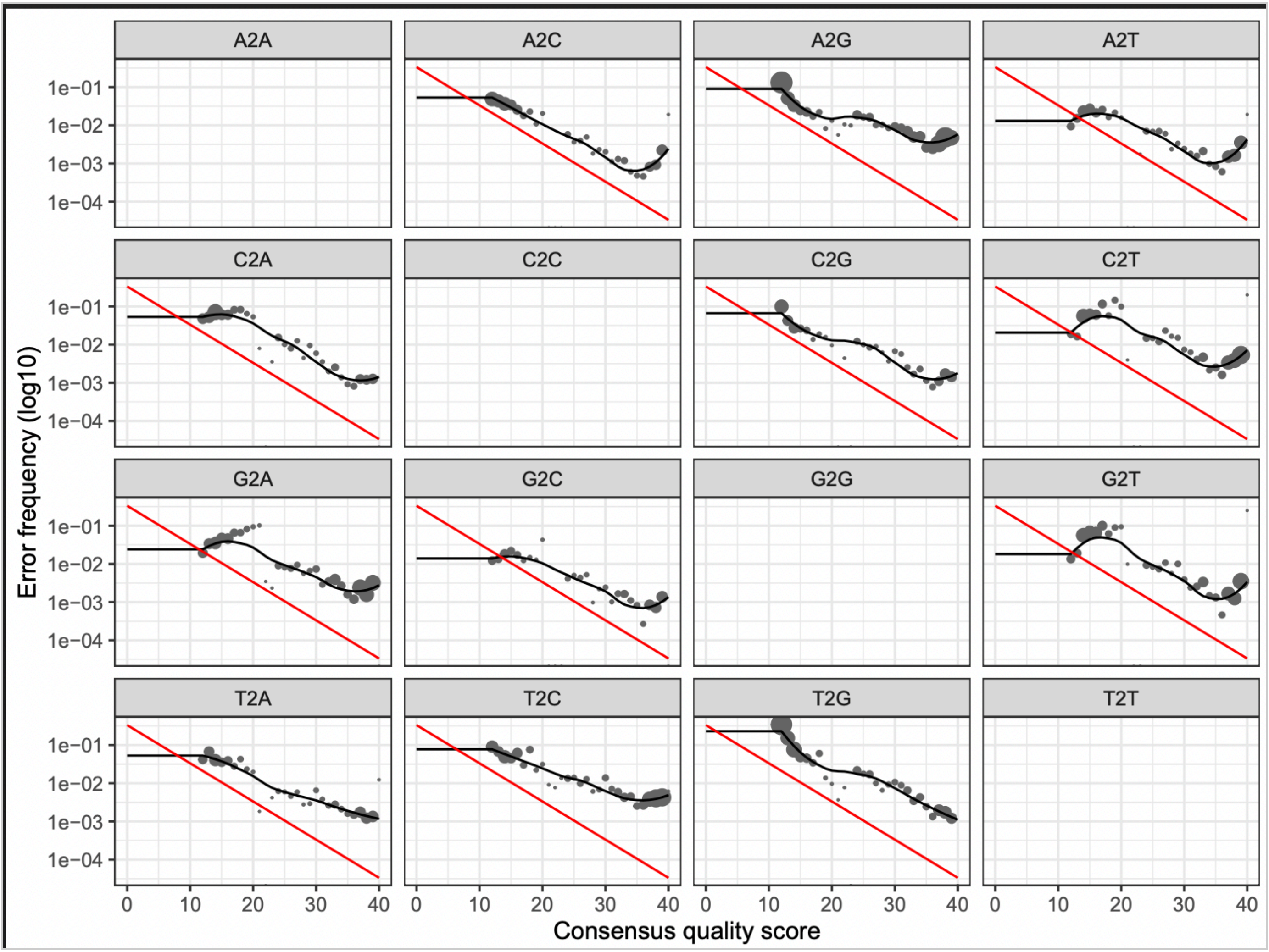

Does the fitted model generally follow the observed error pattern?

Somewhat. You can see that the fitted model starts high and ends low, but it's not the straitest nor is it following

Why is learning error rates important before denoising?

It helps the program correct any mistakes done by the machine, thus helping us prevent the removal of too much info.

Part 4 — dada (denoising)

What dada does:

This is the denoising step. DADA2 uses the learned error model to correct likely sequencing errors and infer exact amplicon sequence variants (ASVs).

What is denoising doing?

Denoising helps clean up the information by removing anything deemed unnecessary so that you have a clear picture of the information youre seeking.

How is ASV inference different from OTU clustering?

ASV is much more specific and "nit-picky", doing more investigation in order to prove a more accurate answer, while OTU is an older model that uses the "good enough" method - basically it uses works by assumption of what the data means.

Part 5 — mergePairs

What mergePairs does:
This step merges the forward and reverse reads for each amplicon so you can reconstruct the full sequence from overlapping read pairs.

What is being merged, and why?

The forward and reverse reads from each amplicon so that the full sequence can be reconstructed from overlapping read pairs.

Why do forward and reverse reads need to overlap?

To make sure that the two sets match and the original DNA fragment could be recreated.

If reads fail to merge, what is one likely reason?

One read could have been too long/short.

Part 6 — makeSequenceTable

What makeSequenceTable does:
This step creates the ASV table. Each row is a sample, each column is an ASV, and each cell contains the number of times that ASV was observed in that sample.

What does this table represent biologically?

The entire microbial community found in all samples.

What are the rows, columns, and cell values in the sequence table?

The rows are the amplicon sequence variants (ASVs), columns are our samples, and the cell values are the counts.

Upload screenshot: sequence table output

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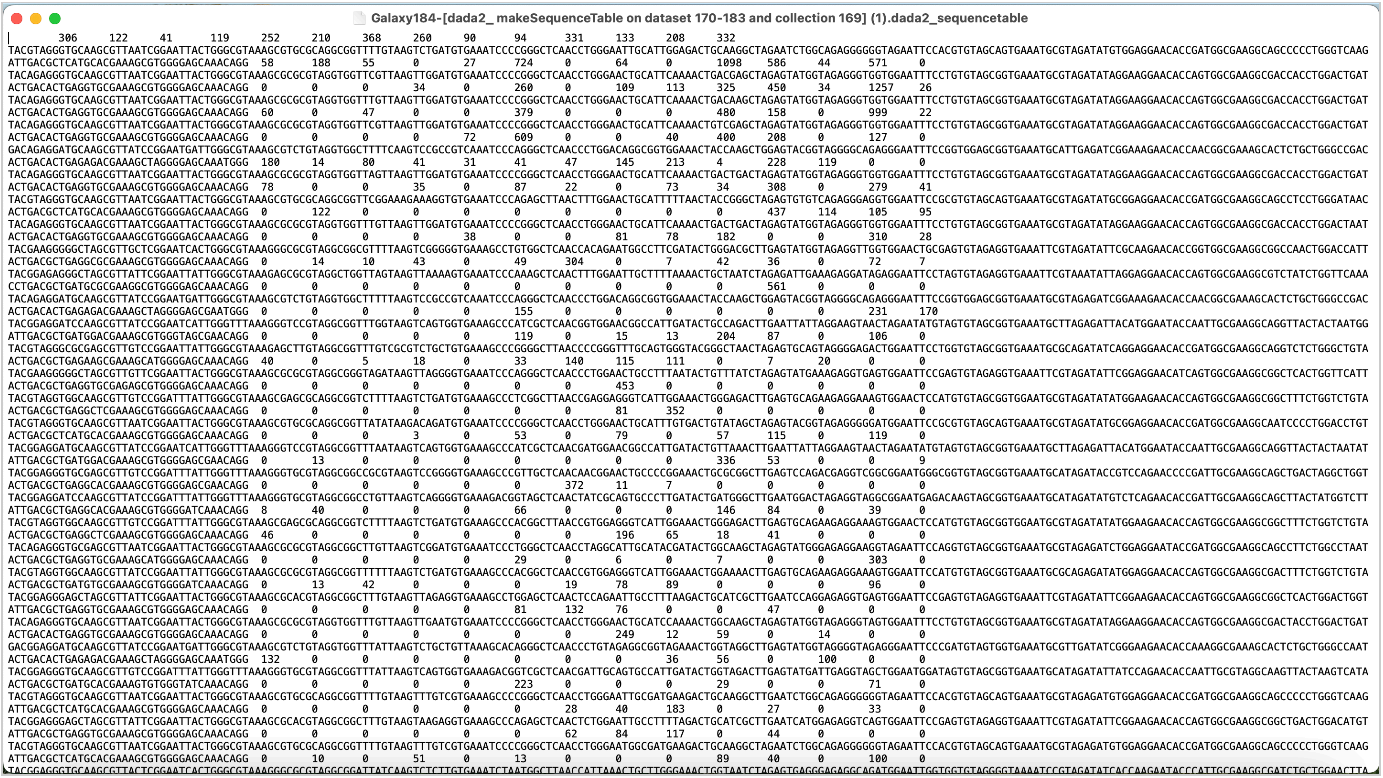

Upload screenshot: sequence length distribution

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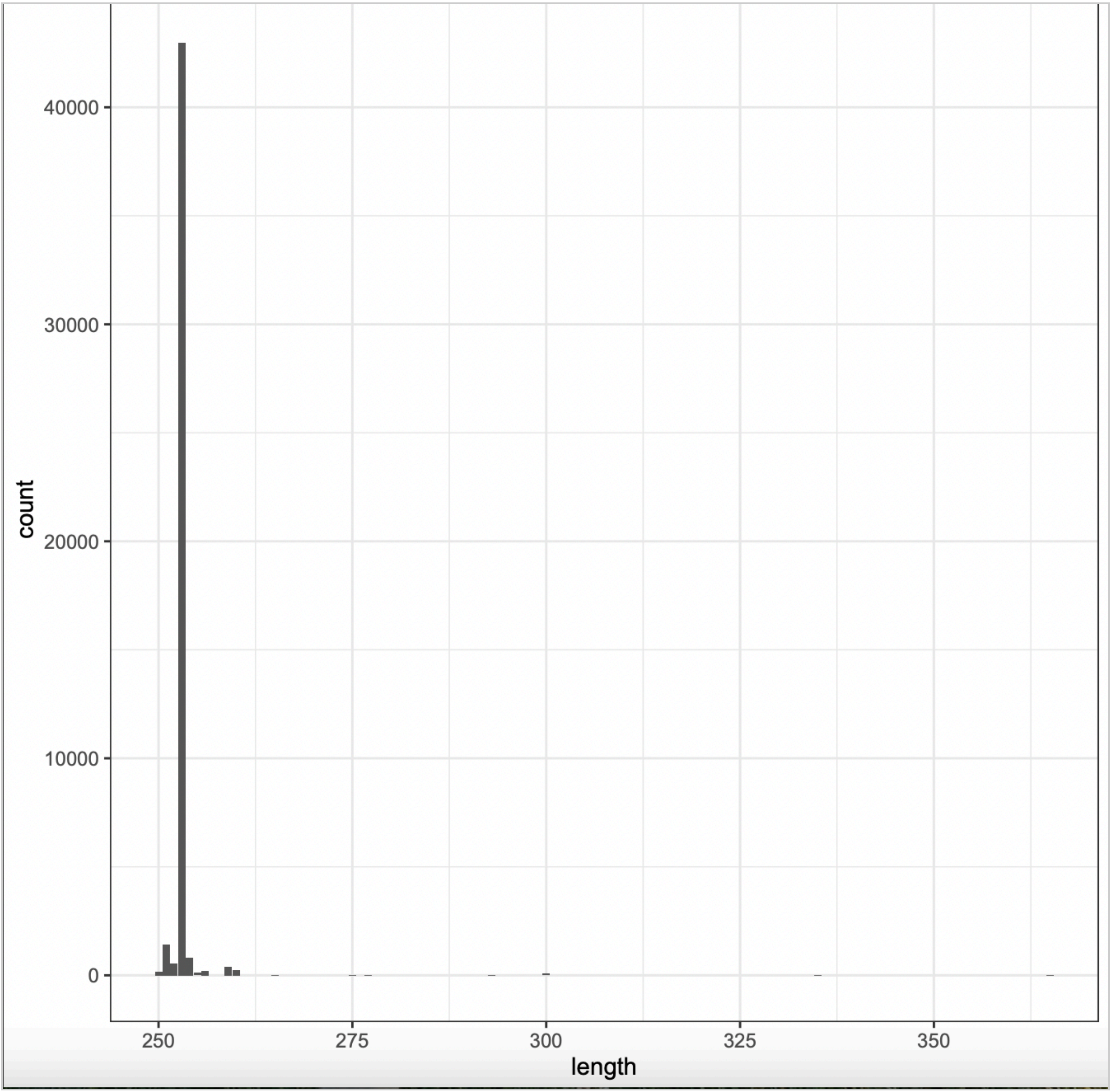

What does the sequence length distribution show?

It helps us be sure that we're looking at real bacterial DNA.

Part 7 — removeBimeraDenovo

What removeBimeraDenovo does:

This step removes chimeras, which are artificial sequences created during PCR when fragments from different templates are incorrectly joined together.

What is a chimera, and why should it be removed?

During PCR, fragments from different templates can sometimes be incorrectly joined together, resulting in an artificial sequence known as a chimera. These chimeras should be removed because they can be misleading

Why are chimeras a problem for interpretation?

As previously mentioned, chimeras can be misleading and lead to incorrect interpretations of results.

How many chimeric reads were removed? How do you know?

10 chimeric reads were removed. Looking back at the "dada2: removeBimeraDenovo" tab, we can see that we started with 731 lines from the input, but the output gave us 721 lines.

Part 8 — assignTaxonomy

What assignTaxonomy does:
This step compares your ASVs to a reference database and assigns taxonomy, such as phylum, class, order, family, genus, or sometimes species.

What does taxonomy assignment add to your ASV table?

Assign Taxonomy tells us what groups our bacteria belongs to, because knowing the groups will allow us to make better assumptions of our bacteria.

Why might some ASVs not be assigned all the way to species?

We could be lacking the information needed in order to assign all the way.

Part 9 — Alpha diversity

9A) Alpha diversity

In your own words, what is alpha diversity?

Alpha diversity is the measurement of the amount of microbial communities within a sample.

9B) Observed richness

What does Observed measure?

Observed richness is the amount of different types of bacteria found.

9C) Shannon diversity

What does Shannon measure?

Shannon measures how "mixed" the group is.

9D) Simpson diversity

What does Simpson measure?

Simpson measures the representation of each group.

Upload screenshot: alpha diversity plot grouped by population

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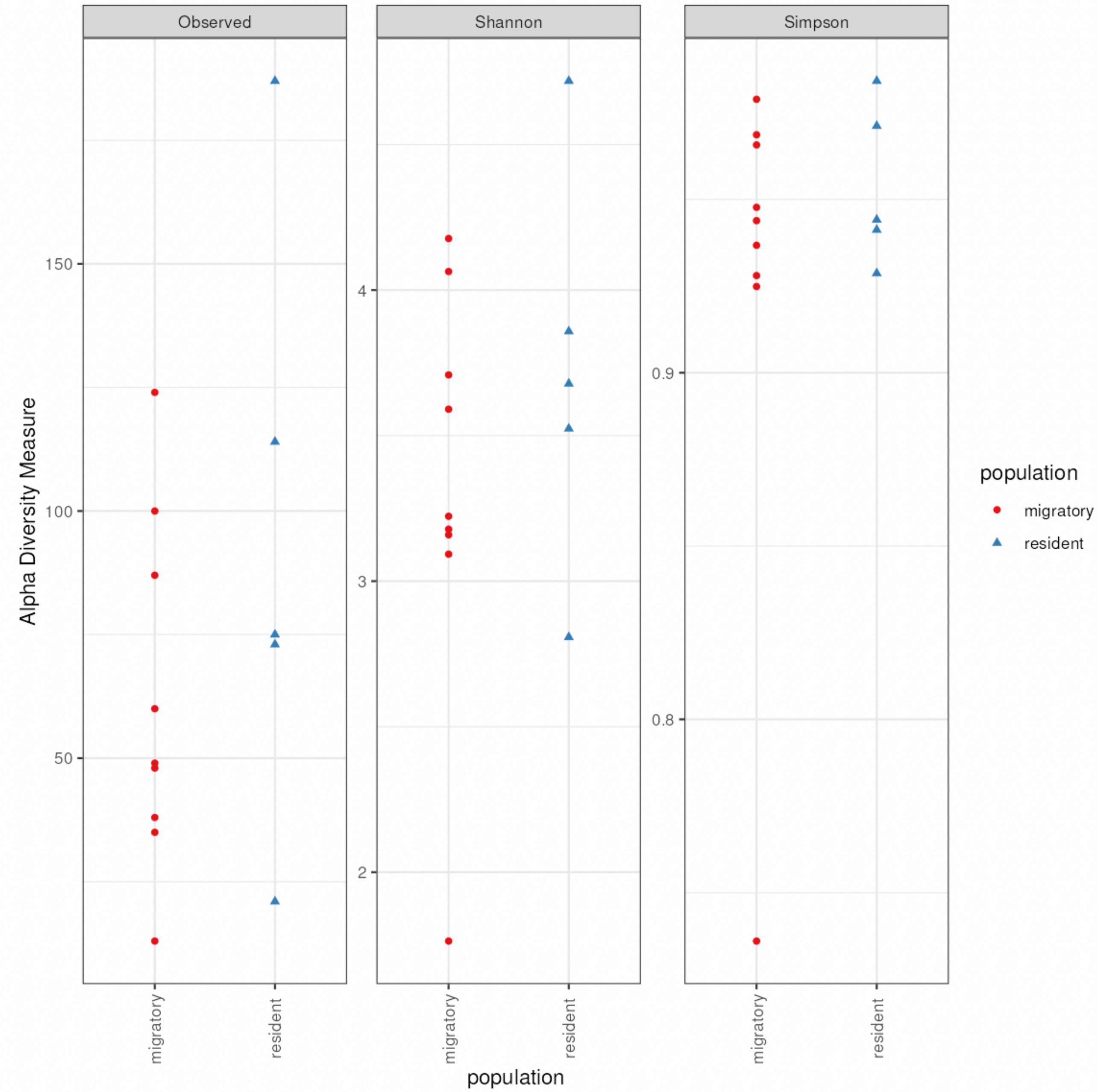

Describe differences in alpha diversity by population.

Observed: The migratory birds tend to stay on the lower end while the resident birds have more of spread
throughout the chart.
Shannon: The migratory birds are more clustered in the mid-top area while the resident birds are more spread

Upload screenshot: alpha diversity plot grouped by sex

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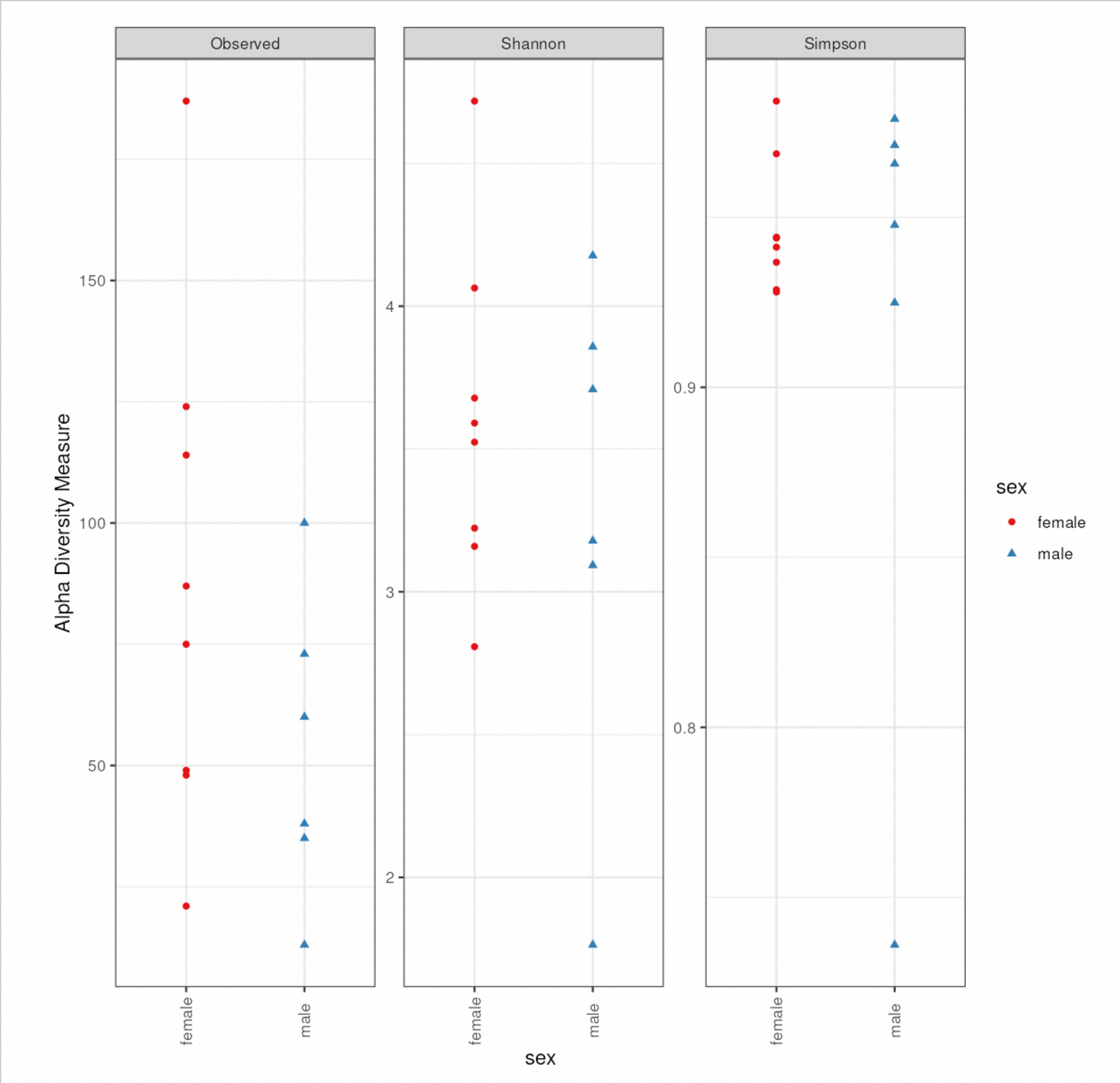

Describe differences in alpha diversity by sex.

Shannon: The females are found on the middle-high end while the males are found organized more along the entire row.

Upload screenshot: alpha diversity plot grouped by flock

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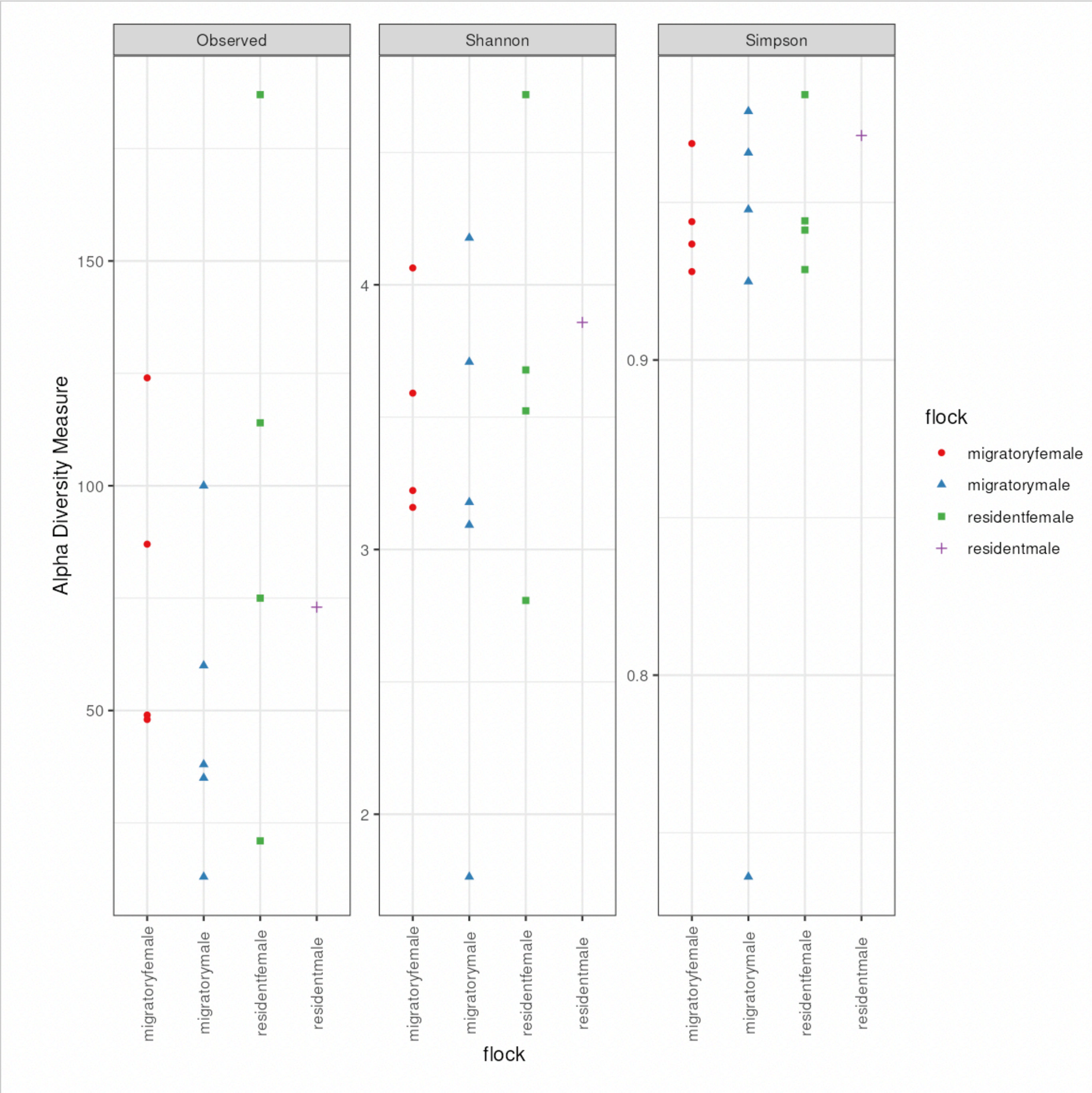

Describe differences in alpha diversity by flock.

Shannon: The migratory males have an outlier found all the way on the bottom while the resident males seem to have a relatively more equal spread.

Part 10 — Beta diversity (NMDS with Bray-Curtis)

What beta diversity means:

Beta diversity describes differences in community composition **between samples.**

In your own words, what is beta diversity?

Beta diversity is the comparison of the diversity between samples.

10A) Bray-Curtis dissimilarity

What does Bray-Curtis measure?

Bray-Curtis measures the amount of bacteria found within a sample.

10B) NMDS

What does NMDS show?

NMDS explains the similarity between samples using distance.

Upload screenshot: NMDS Bray-Curtis plot colored by population

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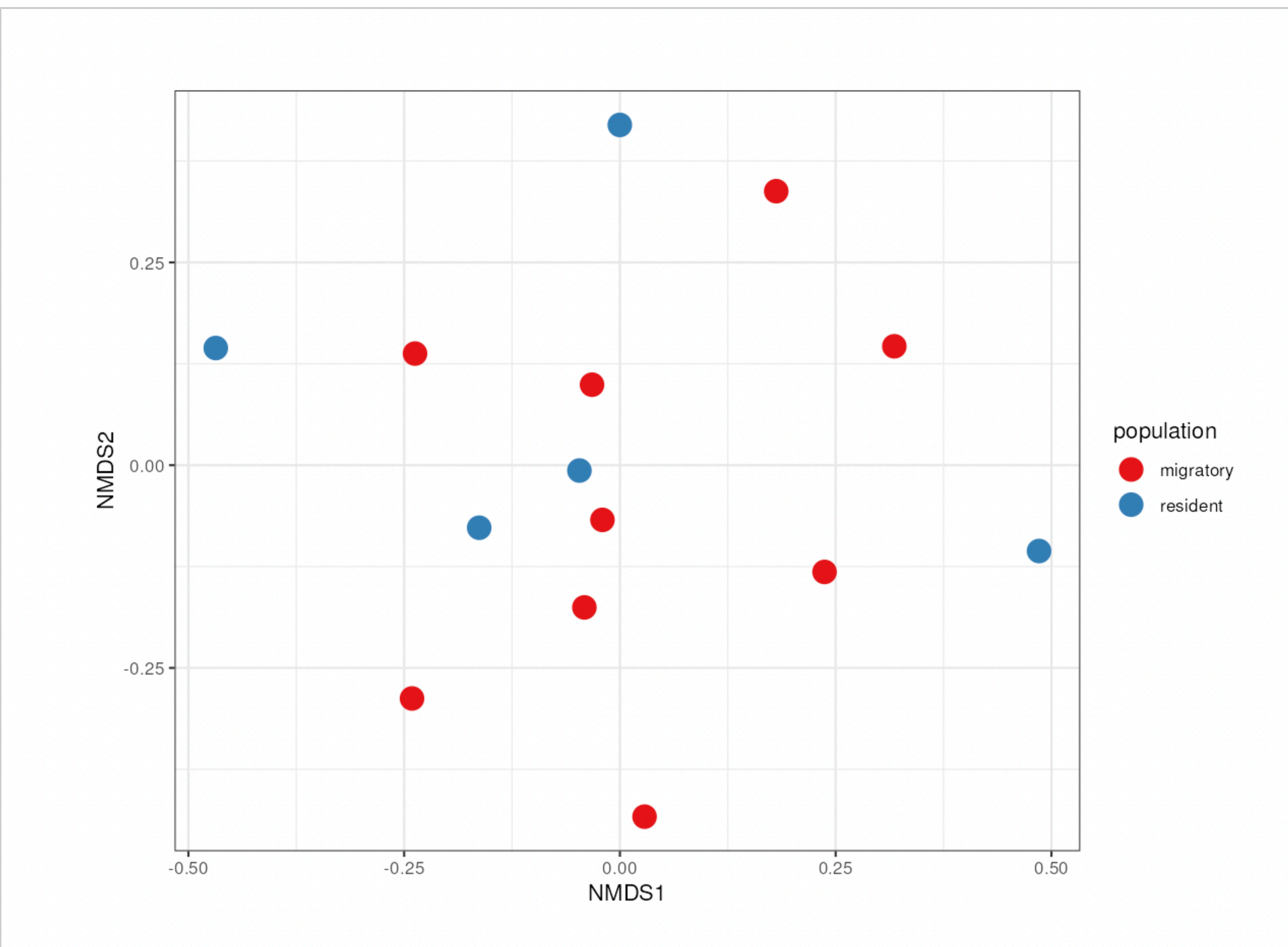

Describe beta diversity patterns by population.

Overall, the points on the Bray-Curtis plot seem to be evenly distributed around the chart. This plot does also make it easier to see the number of differences between the two samples.

Upload screenshot: NMDS Bray-Curtis plot colored by sex

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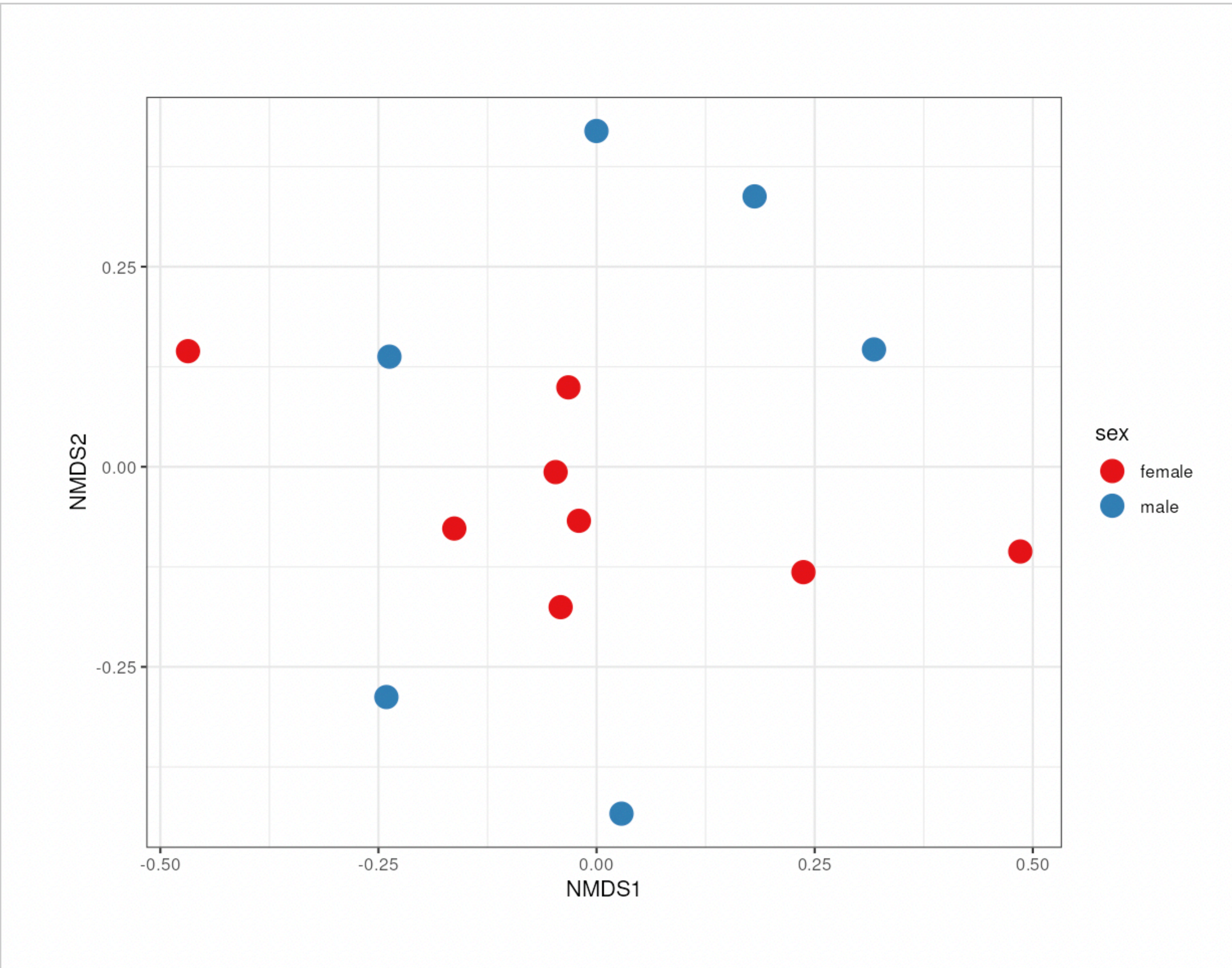

Describe beta diversity patterns by sex.

For the most part, there is no major clustering between the sexes but, you can see some slight clustering in the center for the female birds. The males seem to be pretty even spread around the chart.

Upload screenshot: NMDS Bray-Curtis plot colored by flock

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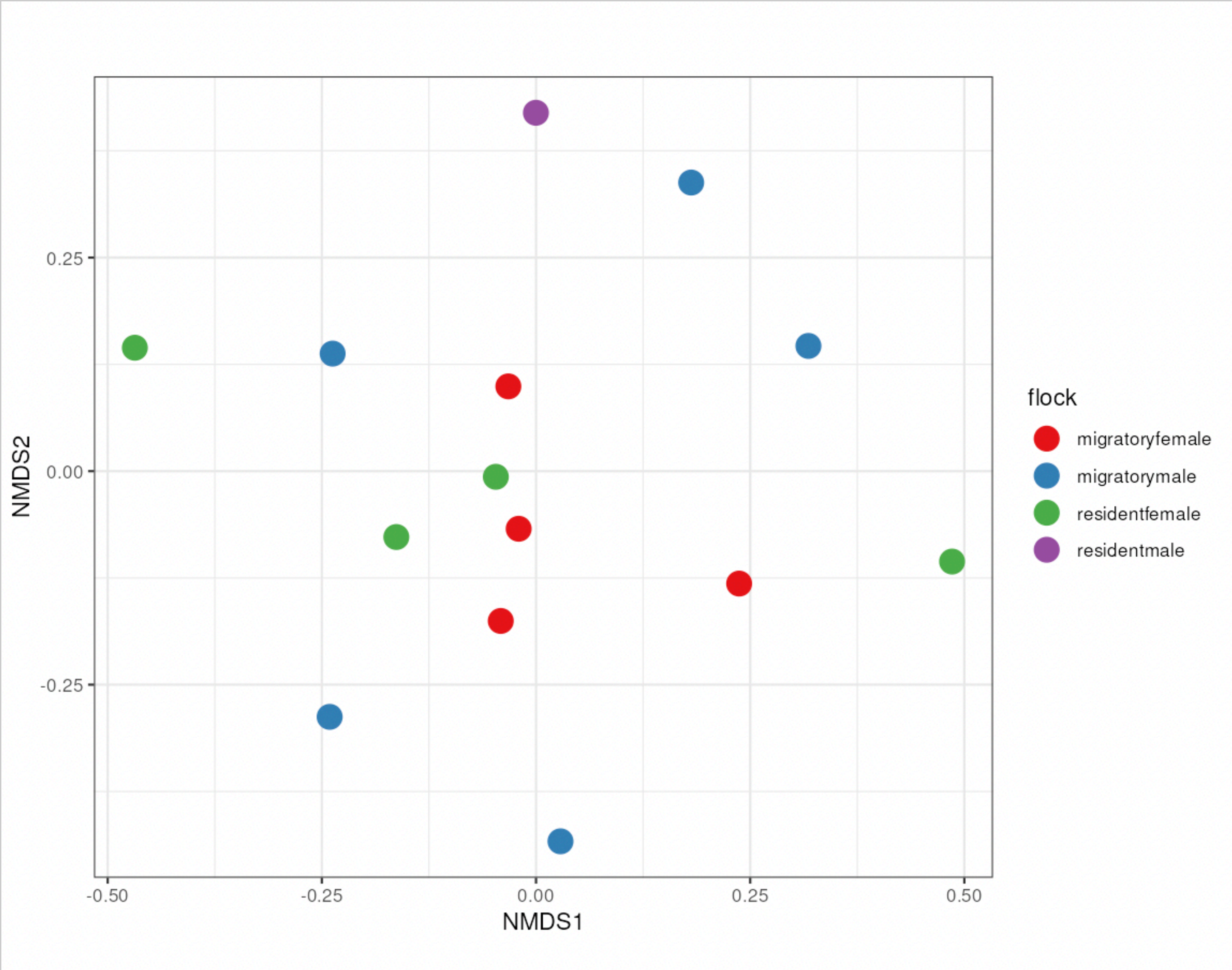

Describe beta diversity patterns by flock.

The flocks seem to be even spread around. The only "maybe" cluster we have is the migratory females as those are found more in the center.

Part 11 — Taxonomic bar plot

What a taxonomic bar plot shows:
A bar plot summarizes the composition of samples by showing the relative abundance of taxa at a selected taxonomic level.

What taxonomic level did you use?

Species

How did you facet the plot?

Not Faceted - would not generate a bar plot if it was.

Why was this taxonomic level useful?

It gave me the most common species that were found.

Why was your faceting choice useful?

I didn't facet because the bar graph would not generate whenever I tried.

Upload screenshot: taxonomic bar plot

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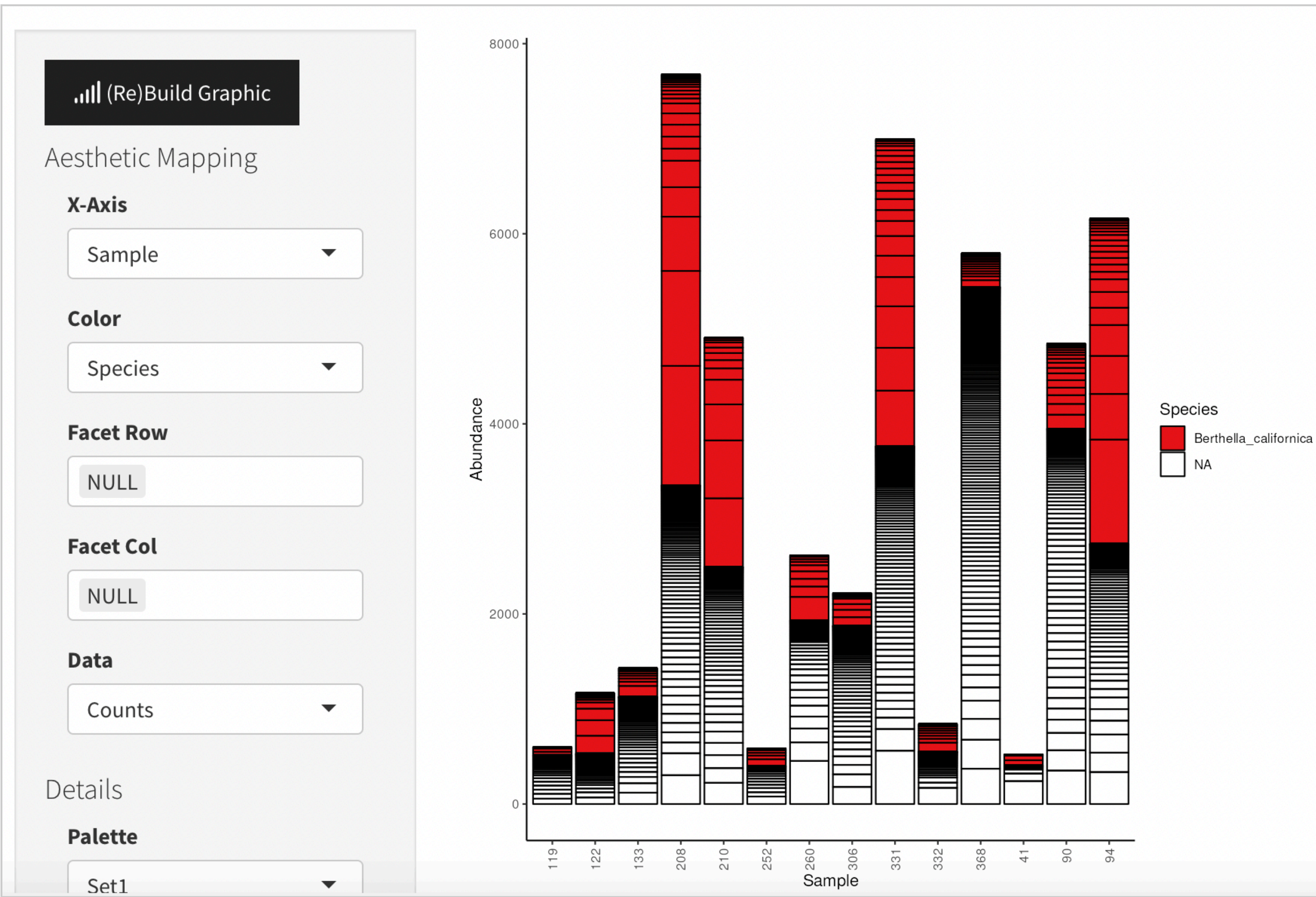

Which taxa appear to be most abundant?

I got something that says Berthella californica but thats a sea slug (a cute one but still not bird germs)...

Did taxonomic composition differ among groups?

I'm not sure. Something went wrong with my data and I ended up getting a cute sea slug as the most prevalent bacterial species :/

Part 12 — Overall interpretation

In 3–5 sentences, summarize what your DADA2 results, alpha diversity, beta diversity, and bar plot together suggest about this dataset.

All in all, the steps followed allowed us figure out what the most prevalent bacteria was in our bird samples. By the end, something may have gone wrong as the bar-graph showed that the most prevalent sample came from a sea slug (Berthella California). Although my work may have messed up somewhere, it is extremely important for researches to know and understand so that they can know how to clean their data in order to receive accurate results.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset
Project/sample set: Bird Microbiome
Sequencing type: Paired-End Sequencing
Grouping variable: Population, Sex, & Flock

Quality profiles
Purpose: To make sure that our sample is clear of any contaminants. Offers a visual of the quality.
Reverse vs forward quality: The reverse read tends to degrade faster, meaning it drops below the 30 mark much sooner than the forward-read quality.
Forward quality drop: around 250
Reverse quality drop: Around 230
Forward truncation: 250
Reverse truncation: 230
Why these values: Usually, most values over 30 are accurate. It's normally when values drop under 30 that the error rate begins to increase.

filterAndTrim
Purpose: The Filter & Trim step cleans the data so that we're left with only high-quality pairs, allowing us to avoid errors.
Parameters used:

```
I only changed my truncate length, which was 250 forward and 230 reverse.
```

If trimmed too much: We might not be able to make the sequences merge into a single sequence.
If not trimmed enough: We could get incorrect results, leading to false assumptions.

learnErrors
Purpose: Learn Errors helps our program learn the "quirks" of our machine. It's important so that we can avoid the mistakes of misidentification of species.
Model fit: Somewhat. You can see that the fitted model starts high and ends low, but it's not the straitest nor is it following the red line perfectly.
Why important: It helps the program correct any mistakes done by the machine, thus helping us prevent the removal of too much info.

dada + mergePairs
Denoising purpose: Denoising helps clean up the information by removing anything deemed unnecessary so that you have a clear picture of the information youre seeking.
ASVs vs OTUs: ASV is much more specific and "nit-picky", doing more investigation in order to prove a more accurate answer, while OTU is an older model that uses the "good enough" method - basically it uses works by assumption of what the data means.
mergePairs purpose: The forward and reverse reads from each amplicon so that the full sequence can be reconstructed from overlapping read pairs.
Need for overlap: To make sure that the two sets match and the original DNA fragment could be recreated.
One reason reads fail to merge: One read could have been too long/short.

Sequence table + sequence length distribution + chimera removal + taxonomy
Sequence table purpose: The entire microbial community found in all samples.
Rows/columns/cells: The rows are the amplicon sequence variants (ASVs), columns are our samples, and the cell values are the counts.
Sequence length distribution: It helps us be sure that we're looking at real bacterial DNA.
Chimera removal purpose: During PCR, fragments from different templates can sometimes be incorrectly joined together, resulting in an artificial sequence known as a chimera. These chimeras should be removed because they can be misleading since they don't actually exist.
Why chimeras are a problem: As previously mentioned, chimeras can be misleading and lead to incorrect interpretations of results.
Taxonomy purpose: Assign Taxonomy tells us what groups our bacteria belongs to, because knowing the groups will allow us to make better assumptions of our bacteria.
Taxonomy limits: We could be lacking the information needed in order to assign all the way.

Alpha diversity
Definition: Alpha diversity is the measurement of the amount of microbial communities within a sample.
Observed: Observed richness is the amount of different types of bacteria found.
Shannon: Shannon measures how "mixed" the group is.
Simpson: Simpson measures the representation of each group.
Differences by population: Observed: The migratory birds tend to stay on the lower end while the resident birds have more of spread throughout the chart. Shannon: The migratory birds are more clustered in the mid-top area while the resident birds are more spread through the center-top. Simpson: In the population graph, both groups are found on the top end but the migratory group has an outliers way down below.
Differences by sex: Observed: The females have a broader spread than the males with an outlier on the top unlike the makes which, although not clustered, are found along the middle-bottom. Shannon: The females are found on the middle-high end while the males are found organized more along the entire row. Simpson: Although both groups are found up high, the males are slightly lower and have an outlier that's found way down below.
Differences by flock: Observed: The resident females tend to be higher overall than the other bird groups while the migratory males are the lowest overall. Shannon: The migratory males have an outlier found all the way on the bottom while the resident males seem to have a relatively more equal spread. Simpson: In this chart, all bird groups are found mainly on top with roughly equal spreads on the top half but the migratory males have an obvious outliers found at the very bottom.

Beta diversity (NMDS Bray-Curtis)
Definition: Beta diversity is the comparison of the diversity between samples.
Bray-Curtis: Bray-Curtis measures the amount of bacteria found within a sample.
NMDS: NMDS explains the similarity between samples using distance.
Patterns by population: Overall, the points on the Bray-Curtis plot seem to be evenly distributed around the chart. This plot does also make it easier to see the number of differences between the two samples.
Patterns by sex: For the most part, there is no major clustering between the sexes but, you can see some slight clustering in the center for the female birds. The males seem to be pretty even spread around the chart.
Patterns by flock: The flocks seem to be even spread around. The only "maybe" cluster we have is the migratory females as those are found more in the center.

Taxonomic bar plot

Taxonomic level: Species
Faceted by: Not Faceted - would not generate a bar plot if it was.
Why this taxonomic level? It gave me the most common species that were found.
Why this faceting choice? I didn't facet because the bar graph would not generate whenever I tried.
Most abundant taxa: I got something that says Berthella californica but thats a sea slug (a cute one but still not bird germs)...
Did composition differ among groups? I'm not sure. Something went wrong with my data and I ended up getting a cute sea slug as the most prevalent bacterial species :/

Overall interpretation

All in all, the steps followed allowed us figure out what the most prevalent bacteria was in our bird samples. By the end, something may have gone wrong as the bar-graph showed that the most prevalent sample came from a sea slug (Berthella California). Although my work may have messed up somewhere, it is extremely important for researches to know and understand so that they can know how to clean their data in order to receive accurate results.